Lab Assessment 5 - Multivariate Analysis

1. Load the dataset shared through e-mail into the R environment. Examine the dataset and identify the numerical and categorical variables.
2. Create a scatter plot involving three variables, where two numerical variables are represented on the X- and Y-axes, and the third variable is represented using color.
3. Create a scatter plot involving four variables, where two numerical variables are represented on the X- and Y-axes, the third variable is represented using color, and the fourth variable is represented using shape.
4. Create a scatter plot involving four variables, where two numerical variables are represented on the X- and Y-axes, the third variable is represented using color, and the fourth numerical variable is represented using point size.
5. Construct a bubble plot using three numerical variables, where two variables are represented on the X- and Y-axes, and the third variable determines the size of the bubbles.
6. Generate subplots using facet_wrap() by selecting an appropriate categorical variable from the dataset.
7. Generate subplots using facet_grid() by selecting two appropriate categorical variables from the dataset. Arrange the plots in rows and columns and compare the patterns observed across the different groups.
8. Construct a pairs plot using at least four numerical variables from the dataset. Use the plot to identify possible relationships and correlations among the selected variables.
9. Create a parallel coordinate plot using at least four numerical variables from the dataset. Differentiate the observations using an appropriate categorical variable.